

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Genes which can mutate to (oncogenes which) cause {uncontrolled cell division/mitosis/cancer}	1			1		
		(ii)		Control of {gene expression/ owtte e.g. transcribed and translated/ switched on or off} by {factors/ named factors} other than changes in DNA sequence.	1			1		
		(iii)		{Undifferentiated/ unspecialised} cell that has the ability to become {specialised/different/ any/ many} cell(s)/ reference to totipotent/ pluripotent	1			1		
	(b)	(i)		X = Transcription Y = Translation Both required for 1 mark	1			1		
		(ii)		Any three × (1) from: A. mRNA (from regulator) not produced(1) B. so regulator protein not produced (1) C. Suppressor gene not {expressed/ switched on}/ suppressor gene remains switched off/ suppressor protein not produced(1) D. Uncontrolled {mitosis/ cell division} (1)			3	3		
		(iii)		Any three × (1) from: A. Remove acetyl groups (from histones). (1) B. (So) DNA becomes tightly coiled.(1) C. RNA polymerase can't access {gene/DNA} (1) D. cannot be transcribed (1)			3	3		
		(iv)		<u>Increased</u> (risk of) cancer/ <u>increased</u> (risk of) tumour formation/ (1) Passed onto next generation. (1)		2		2		
				Question 4 total	4	2	6	12	0	0